

Embodied Miracle-Question Counseling in Virtual Reality

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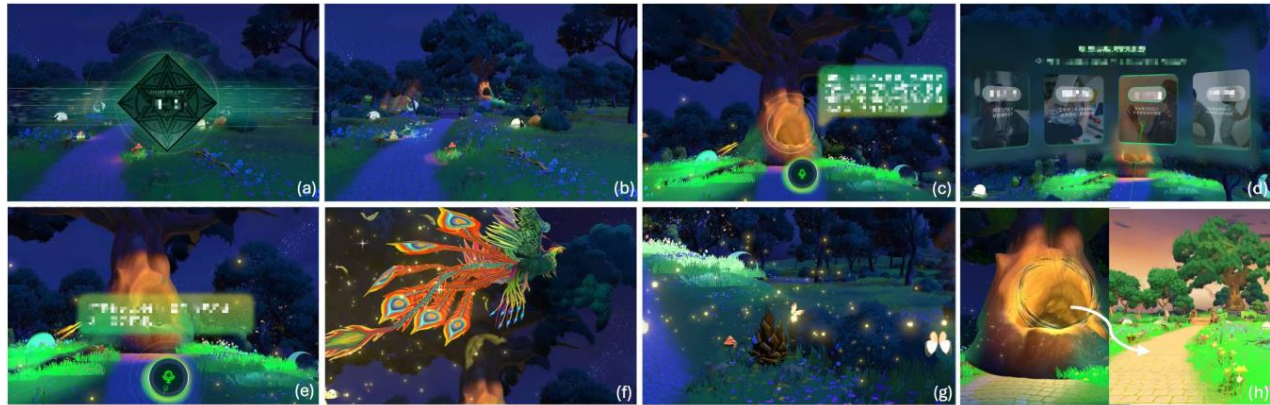


Figure 1: VR-based embodied Miracle Question (MQ) interaction flow. (a–b) Entry into a non-anthropomorphic, low-arousal virtual environment with path-based guidance for attentional focusing and psychological immersion. (c) Arrival at a non-anthropomorphic interaction space centered on a virtual tree, where the script-constrained, state-aware MQ dialogue process is initiated. (d) Selection of a reflection topic to initialize the dialogue state machine. (e) Visual presentation of dialogue prompts and user inputs governed by predefined scripts and dialogue states, with a large language model used only for natural language generation. (f) A dialogue-state-triggered embodied “miracle moment” with non-anthropomorphic visual effects and low-intensity haptic feedback. (g) Continuous adaptation of environmental lighting and natural element motion to dialogue states as low-level, non-semantic embodied feedback. (h) Transition to a brighter natural environment marking session completion.

ABSTRACT

The Miracle Question (MQ) is a core technique in Solution-Focused Brief Therapy (SFBT) that guides individuals to envision a future in which their current problems no longer exist, supporting goal-oriented reflection and action planning. In practice, however, MQ relies heavily on users’ imagination and verbal articulation, which can limit engagement and increase cognitive load. To address this limitation, we present a virtual reality (VR)–based embodied counseling system that operationalizes the structured MQ process within an immersive, non-anthropomorphic environment. The system formalizes MQ as a script-constrained, state-aware dialogue model, in which predefined dialogue states govern the counseling flow while a large language model is used solely for natural language generation. Instead of employing a humanoid virtual agent, the system adopts a nature-inspired interaction metaphor designed to minimize social evaluative cues and support self-projection. A key design element is an embodied miracle moment, in which the imagined future state is externalized

as a symbolic, multisensory VR event, transforming abstract verbal imagination into a perceptible embodied experience. In a pilot pre-post evaluation, participants showed increased positive affect and reduced negative affect after a single MQ-guided VR session. In a separate between-subject comparison against a humanoid-agent-based presentation, the non-anthropomorphic condition yielded higher perceived psychological safety and greater self-disclosure. These results suggest that VR can provide an effective embodied pathway for structured reflective counseling, demonstrating how controlled dialogue flow and environment-driven embodiment can be integrated in automated reflective systems without real-time human intervention.

Index terms: Virtual reality, Miracle Question, embodied interaction, structured reflective support, non-anthropomorphic interaction

1 INTRODUCTION

Access to psychological support remains limited worldwide due to shortages of trained professionals, high service costs, and persistent social stigma [1]. Although technology-mediated interventions have been explored to improve accessibility and scalability [2], many psychological support practices still rely on synchronous, face-to-face, language-centered interaction, which constrains everyday availability and large-scale deployment. Solution-Focused Brief Therapy (SFBT) offers a structured and goal-oriented alternative that emphasizes future-oriented meaning-

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making rather than problem analysis [3]. The Miracle Question (MQ), a core technique within SFBT, guides individuals to imagine a future in which their current problems have been resolved and to reflect on the changes and actions that would make such a future possible. Prior studies indicate that MQ can support hope, self-efficacy, and action-oriented reflection [4]. However, the effectiveness of MQ depends strongly on how the dialogue is facilitated and on clients' ability to engage imaginatively with future scenarios, making the technique sensitive to interaction design and guidance strategies [5].

Virtual reality (VR) provides a means to externalize internally imagined processes into perceptible and immersive experiences. Prior VR-based mental health systems demonstrate that immersive environments can enhance experiential engagement; however, these systems primarily emphasize exposure, simulation, or skills training and seldom encode structured reflective dialogue grounded in established counseling techniques [6]–[9]. In parallel, recent advances in large language models enable automated dialogue systems for psychological support [10], [11], but these systems remain predominantly text- or voice-based and provide limited embodied support beyond conversation.

In this work, we present a VR-based system that operationalizes the Miracle Question within an embodied interactive environment. Rather than simulating a human counselor, the system adopts a non-anthropomorphic, nature-inspired interaction paradigm designed to reduce social evaluative cues and support self-directed reflection. The MQ process is formalized as a script-constrained, state-aware dialogue model, in which dialogue flow is governed by predefined states while a large language model is used solely for natural language generation. A key design feature is an embodied miracle moment, where future-oriented imagination is externalized as a symbolic multisensory VR event. Through a pilot study comparing non-anthropomorphic and humanoid-agent interaction conditions, we examine how interaction style influences emotional outcomes, psychological safety, and self-disclosure during a single MQ session.

2 RELATED WORK

The MQ was introduced by de Shazer as a central technique within Solution-Focused Brief Therapy and has been widely applied in counseling and social work practice [3]. Empirical studies by Gingerich and Peterson [4] show that MQ can support hope, self-efficacy, and action-oriented reflection. Subsequent analyses of therapeutic dialogue suggest that the effectiveness of MQ is influenced by how reflective imagination is supported through ongoing interaction, as well as by clients' responsiveness and engagement during the questioning process [5]. As a result, MQ is most commonly implemented through face-to-face, language-centered interaction, where practitioners dynamically adapt questioning strategies to clients' responses.

VR has been extensively studied in mental health interventions, particularly in exposure-based therapy and emotion regulation. Representative systems such as BraveMind by Rizzo et al. [6] demonstrate that immersive environments can enhance emotional engagement compared to imagination-based approaches. Freeman et al. [7] further show that VR can improve experiential involvement across a range of mental health applications. However, most VR-based systems emphasize scenario simulation or behavioral rehearsal, and rarely encode the structured process of specific counseling techniques, such as staged reflective questioning.

In parallel, conversational agents and large language models have enabled automated psychological support systems. Fitzpatrick et al. [10] demonstrate the feasibility of a fully automated conversational agent for delivering CBT. Koulouri et al. [11] report an exploratory study on the acceptability of mental-health chatbots among young

adults. More recently, Shimojo et al. [12] integrate a solution-focused counseling approach into a metaverse environment using a humanoid virtual agent, where counseling is primarily realized through dialogue and the virtual environment serves as a setting for social interaction.

In contrast, our work does not center on replacing the counselor with a virtual human agent. Instead, we treat the immersive environment itself as the primary interaction medium and focus on how non-anthropomorphic, nature-inspired environments can support self-directed reflection by reducing social evaluative cues. The Miracle Question is operationalized as a structured interaction flow, in which key reflective stages are externalized as embodied environmental transitions rather than remaining purely linguistic.

3 PILOT SYSTEM

We present a virtual reality-based embodied system that operationalizes the MQ as a structured, experience-centered interaction process executable without real-time human facilitation. Script-constrained, state-aware dialogue control is integrated with environment-driven embodiment to ensure that reflective interaction remains stable and repeatable under automated execution. Figure 2 provides an overview of the system architecture.

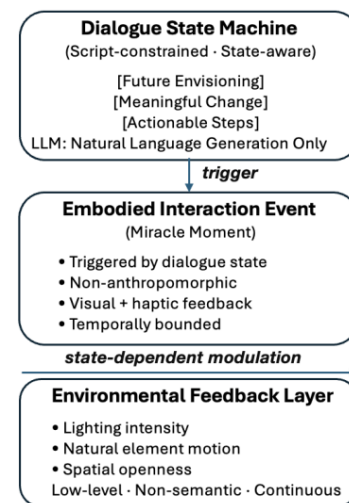


Figure 2: System architecture for dialogue-driven embodied interaction.

3.1 Script-Constrained and State-Aware Dialogue Model

To support reliable execution of the Miracle Question in an automated VR setting, the counseling process is formalized as a script-constrained, state-aware dialogue model grounded in its standard formulation within Solution-Focused Brief Therapy. The MQ process is implemented as a finite set of dialogue states corresponding to its core phases: future envisioning, identification of meaningful change, and reflection on actionable steps.

During runtime, user input is continuously interpreted with respect to the active dialogue state. Each state specifies a bounded set of reflective objectives, prompt templates, and allowable state transitions. When user responses satisfy predefined semantic or cognitive criteria—such as articulating a concrete future scenario or identifying a specific change—the system advances to the next state. If these conditions are not met, the system remains in the current state and adapts subsequent prompts to maintain reflective focus without prematurely progressing the dialogue.

Within this control framework, the large language model is used exclusively for natural language generation and contextual adaptation. The dialogue script determines what is asked and when transitions occur, while the language model generates utterances aligned with the reflective goal of the current state. This separation prevents uncontrolled generative drift while preserving conversational naturalness.

To address expressive difficulty or cognitive stagnation, the system incorporates state-aware guidance modulation. When user responses are vague, repetitive, or emotionally diffuse, question granularity is reduced by focusing on specific situations or small experiential changes. As user descriptions become more concrete, prompts shift toward translating reflection into actionable steps. All dialogue is framed in a non-evaluative and non-role-based manner, avoiding authoritative or judgmental cues.

3.2 Virtual Therapeutic Environment Design

The virtual environment is designed as an integral component of the MQ interaction flow rather than as a representation of a real counseling room. Its primary role is to function as an experiential scaffold for reflective counseling by providing a stable, low-arousal, and non-directive perceptual context that supports sustained introspection alongside dialogue progression.

The visual forms of selected environmental elements draw loosely from natural and myth-inspired imagery, used solely to support non-anthropomorphic perception rather than to convey cultural narratives. As illustrated in Figure 1, the environment adopts a nature-inspired spatial structure centered on a virtual tree and its interior space, with a tree hollow serving as the primary interaction medium. This non-anthropomorphic design minimizes social evaluative cues, interpersonal role expectations, and implicit authority commonly associated with counselor-led interaction, allowing users to maintain an introspective stance and direct attention toward their own experience and imagined future.

Ambient elements such as lighting and natural motion function as continuous perceptual feedback synchronized with the dialogue state. As the MQ process advances, these elements gradually shift in brightness, density, and movement dynamics, providing implicit cues of progression without introducing explicit semantic content or task demands.

Throughout the session, the environment remains passive and non-instructive, avoiding explicit behavioral guidance or evaluative feedback. In this way, verbal reflection remains central, while embodied perception provides a continuous and supportive experiential backdrop for future-oriented imagination.

3.3 Embodied Interaction for the Miracle Moment

Embodied interaction is most explicitly manifested during the miracle moment, the phase in which users are guided to experience a future state where the presenting problem is resolved. This phase is implemented as a dialogue-triggered embodied interaction event that externalizes future-oriented imagination into a perceptible experiential transition.

Concretely, the miracle moment is instantiated through a temporally bounded, particle-based interaction event, in which a spatially distributed visual field and low-intensity controller-based haptic feedback jointly signal the transition into the resolved future state. This interaction event is strictly synchronized with the active dialogue state and does not introduce additional semantic content beyond marking the experiential shift.

During this phase, a non-anthropomorphic virtual entity functions as a perceptual anchor to indicate the initiation and completion of the miracle moment. The entity does not engage in dialogue or convey symbolic meaning, but serves to structure the temporal boundaries of the interaction without introducing social role cues or authoritative presence.

Upon completion of the embodied interaction event, the system transitions back to reflective dialogue, prompting users to articulate actionable steps grounded in the experienced future state. In parallel, gradual changes in environmental lighting and natural motion continue to provide low-level embodied feedback across the session, implicitly mirroring shifts in reflective state without requiring explicit attention or interpretation from the user.

4 EVALUATION

4.1 Experimental Design

We conducted a two-study pilot evaluation of the proposed VR MQ system.

Study A employed a within-subject pre-post design to examine short-term affective changes induced by the proposed non-anthropomorphic VR MQ system. Ten participants ($n = 10$) completed a single 20–25-minute MQ-guided VR session and filled out the Positive and Negative Affect Schedule (PANAS) immediately before and after the session. System usability was assessed using the System Usability Scale (SUS).

Study B employed a between-subject design to compare two presentation variants of the same system: a non-anthropomorphic environment-based interaction (NA) and a humanoid-agent-based interaction (HA). Twenty participants who did not take part in Study A were randomly assigned to one of the two conditions (NA: $n = 10$; HA: $n = 10$). The MQ dialogue structure, content, and timing were kept identical across conditions, with only the interaction presentation differing. In the NA condition, dialogue prompts were presented through environment-centered, non-anthropomorphic interaction, whereas in the HA condition the same prompts were delivered by a humanoid virtual agent. Participants completed PANAS immediately before and after the session; they also completed post-session ratings of psychological safety (PS), perceived judgment (PJ), and self-disclosure (SD) on 7-point Likert scales (1–7).

All sessions were conducted in a quiet VR laboratory using a Pico Neo3 Pro Eye headset with 6DoF head and hand tracking.

Paired-sample t -tests were used for pre-post comparisons in Study A, and independent-samples t -tests were used for between-condition comparisons in Study B, with a significance level of $\alpha = 0.05$. This study was conducted as a non-clinical pilot for reflective support; the system does not provide diagnosis or treatment, and all participants provided informed consent.

4.2 Results

4.2.1 Emotional Effects and Usability of the VR MQ System (Study A)

All participants in Study A completed the experimental procedure. Descriptive statistics of PANAS scores before and after the MQ-guided VR session are summarized in Table 1. Prior to the intervention, participants reported a mean positive affect (PA) score of 25.5 ($SD = 2.5$) and a mean negative affect (NA) score of 20.0 ($SD = 2.0$). After the session, mean PA increased to 30.0 ($SD = 2.0$), while mean NA decreased to 17.5 ($SD = 1.5$).

Paired-sample t -tests ($df = 9$) revealed a significant increase in positive affect ($t = 10.0, p < 0.001$) and a significant decrease in negative affect ($t = -8.0, p < 0.001$). These results indicate that a single use of the proposed non-anthropomorphic VR MQ system can produce statistically significant short-term emotional changes. System usability was also evaluated in Study A using the System Usability Scale (SUS). The mean SUS score was 78.3 ($SD = 8.7$), with a Cronbach's α of 0.89, indicating good internal consistency. According to established SUS interpretation guidelines, this score corresponds to a "Good" level of usability.

Table 1. Pre–Post PANAS Scores After Using the Proposed VR MQ System (Study A, $n = 10$)

	Pre-session Mean (<i>SD</i>)	Post-session Mean (<i>SD</i>)	<i>t</i> (<i>df</i> = 9)	<i>p</i>
PA	25.5 (2.5)	30.0 (2.0)	10.0	< 0.001
NA	20.0 (2.0)	17.5 (1.5)	–8.0	< 0.001

4.2.2 Comparison Between Interaction Conditions (Study B)

All participants in Study B completed the assigned interaction condition. Study B examined whether interaction presentation influenced reflective outcomes when the MQ dialogue structure, content, and timing were held constant. As shown in Table 2, participants in the non-anthropomorphic environment-based condition (NA) exhibited larger affective changes than those in the humanoid-agent-based condition (HA).

Specifically, the NA condition demonstrated greater increases in positive affect (Δ PA) and greater reductions in negative affect (Δ NA) compared to the HA condition. Δ PA and Δ NA represent pre–post PANAS change scores within each condition. In addition, participants in the NA condition reported significantly higher psychological safety (PS), lower perceived judgment (PJ), and greater self-disclosure (SD) than participants interacting with a humanoid virtual agent. Statistical details are summarized in Table 2.

These findings suggest that reducing anthropomorphic social cues and treating the immersive environment itself as the primary interaction medium can facilitate a safer and more open reflective experience, even when counseling content and structure remain identical.

Table 2. Comparison Between Interaction Conditions (Study B; NA: $n = 10$, HA: $n = 10$)

	NA Condition Mean (<i>SD</i>)	HA Condition Mean (<i>SD</i>)	<i>p</i>
Δ PA	+4.6 (1.4)	+2.8 (1.6)	0.018
Δ NA	–2.7 (1.1)	–1.4 (1.3)	0.031
PS	6.1 (0.6)	4.8 (0.9)	< 0.01
PJ	2.4 (0.7)	4.2 (0.8)	< 0.001
SD	5.9 (0.7)	4.6 (0.8)	< 0.01

4.3 Discussion

While traditional counseling typically involves a human counselor, the proposed system adopts a non-anthropomorphic design approach in which counseling functions are realized without simulating a human counselor. Instead, the core listening and guiding functions of counseling are operationalized through a script-constrained, state-aware dialogue model tightly coupled with environment-driven embodied transitions. In this configuration, dialogue provides structured cognitive guidance, while embodied environmental change provides non-verbal, non-evaluative experiential feedback, together supporting reflective engagement without interpersonal interaction.

In addition, by operationalizing structured counseling functions without requiring real-time professional presence, the system may help extend access to structured reflective support in contexts where trained counselors are scarce or unavailable.

Against this background, the results provide initial evidence that operationalizing the MQ as an embodied VR interaction shows

promise in supporting short-term reflective counseling without real-time human facilitation. By externalizing future-oriented imagination into a perceptible experience, the proposed system may help alleviate some of the cognitive demands associated with purely verbal reflection and sustained imagination.

The comparison between interaction conditions further indicates that interaction framing plays a critical role in reflective engagement. Although both conditions employed identical dialogue structure and content, the non-anthropomorphic environment-based interaction yielded higher psychological safety and greater self-disclosure than the humanoid-agent-based presentation. This suggests that minimizing anthropomorphic social cues and treating the immersive environment itself as the primary interaction medium may better support self-directed reflection.

In addition to the quantitative results, brief post-session feedback via open-ended questions collected at the end of the experiments indicated that participants generally perceived the non-anthropomorphic VR environment as psychologically safe and non-judgmental, reporting reduced pressure compared to human- or agent-mediated interaction. No participant reported notable VR-related discomfort or cybersickness. These observations are reported descriptively to complement the quantitative findings.

These findings should be interpreted in light of the study’s short-term scope, non-clinical sample, and limited sample size. Future work will incorporate more systematic qualitative analysis and longitudinal evaluation to further examine user experience, embodied comfort, and reflective outcomes over repeated use.

5 CONCLUSION

This paper presented a VR-based embodied system that operationalizes the Miracle Question as a structured, experience-centered interaction without requiring real-time human facilitation. By combining a script-constrained, state-aware dialogue model with a non-anthropomorphic immersive environment, the system enables future-oriented reflection beyond language-centered interaction. Pilot results suggest that embodied, non-anthropomorphic VR interaction can provide an effective and scalable pathway for structured reflective counseling techniques such as the Miracle Question.

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